

Protective Effect of Intestinal *Blautia* Against Neutropenic Fever in Allogeneic Transplant Recipients

Armin Rashidi,^{1,*} Jonathan U. Peled,² Maryam Ebadi,¹ Tauseef Ur Rehman,¹ Heba Elhusseini,¹ LeeAnn T. Marcello,² Hossam Halaweish,³ Thomas Kaiser,³ Shernan G. Holtan,¹ Alexander Khoruts,⁴ Daniel J. Weisdorf,¹ and Christopher Staley^{3,5}

¹Division of Hematology, Oncology, and Transplantation, Department of Medicine, University of Minnesota, Minneapolis, Minnesota, USA; ²Adult Bone Marrow Transplantation Service, Memorial Sloan Kettering Cancer Center and Weill Cornell Medical College, New York, New York, USA; ³Department of Surgery, University of Minnesota, Minneapolis, Minnesota, USA; ⁴Division of Gastroenterology, Hepatology, and Nutrition, Department of Medicine, University of Minnesota, Minneapolis, Minnesota, USA; and ⁵BioTechnology Institute, University of Minnesota, St Paul, Minnesota, USA

Background. Neutropenic fever (NF) occurs in >70% of hematopoietic cell transplant (HCT) recipients, without a documented cause in most cases. Antibiotics used to prevent and treat NF disrupt the gut microbiota; these disruptions predict a higher posttransplantation mortality rate. We hypothesized that specific features in the gut microbial community may mediate the risk of NF.

Methods. We searched a large gut microbiota database in allogeneic HCT recipients (12 546 stool samples; 1278 patients) to find pairs with NF (cases) versus without NF (controls) on the same day relative to transplantation and with a stool sample on the previous day. A total of 179 such pairs were matched as to the underlying disease and graft source. Several other important clinical variables were similar between the groups.

Results. The gut microbiota of cases on the day before NF occurrence had a lower abundance of *Blautia* than their matched controls on the same day after transplantation, suggesting a protective role for *Blautia*. Microbiota network analysis did not find any differences in community structure between the groups, suggesting a single-taxon effect. To identify putative mechanisms, we searched a gut microbiome and serum metabolome database of patients with acute leukemia receiving chemotherapy and identified 139 serum samples collected within 24 hours after a stool sample from the same patient. Greater *Blautia* abundances predicted higher levels of next-day citrulline, a biomarker of total enterocyte mass.

Conclusions. These findings support a model in which *Blautia* protects against NF by improving intestinal health. Therapeutic restoration of *Blautia* may help prevent NF, thus reducing antibiotic exposures and transplantation-related deaths.

Keywords. *Blautia*; hematopoietic cell transplantation; metabolomics; microbiota; neutropenic fever.

In most patients undergoing allogeneic hematopoietic cell transplantation (HCT), neutropenic fever (NF) develops in the early posttransplantation period [1, 2], resulting in prolonged and costly hospitalization, heavy antibiotic exposure, and compromised quality of life. The multitude of antibiotics used to prevent and treat NF in these patients disrupts gut microbial communities [3, 4]; reduced gut microbial diversity has been associated with increased posttransplantation mortality rates [5]. Translocation of bacteria from the gut in the setting of transplant conditioning–related gut barrier damage is a known cause of bacteremia and NF [6–9]. However, because blood cultures and other infectious workup usually fail to find a culprit, the cause of NF remains unknown in most cases

[1]. A better understanding of NF pathogenesis could help identify novel targets for prevention and treatment, leading to improved outcomes after HCT. We have found preliminary evidence for a more complex interaction between the gut microbiota and the host as a mechanism for NF. Specific bacteria can influence the integrity and function of the intestinal epithelium, thus altering the risk of translocation of other microbes and also modulating metabolic cross-talk between the host and microbiota [10, 11].

The study of the gut microbiota in HCT recipients is challenging owing to the high-risk nature of most experimental interventions in this patient population and also the large interindividual variability of clinical factors that may influence both the microbiota and transplant outcomes. In observational studies, the patient cohort should ideally be homogeneous in relevant covariates. Unfortunately, most HCT microbiota databases are not large enough to permit selection of a homogenous cohort with an adequate sample size for powerful analysis. We used a large database of fecal microbiota profiles of allogeneic HCT recipients (12 546 longitudinal stool samples from 1278 patients) to characterize gut microbiota composition within

Received 22 February 2022; editorial decision 8 April 2022; published online 18 April 2022

Correspondence: Armin Rashidi, MD, PhD 14-100 PWB, MMC 480, 420 Delaware St. SE, Minneapolis, MN 55455, Email: arashidi@umn.edu

Clinical Infectious Diseases® 2022;75(11):1912–20

© The Author(s) 2022. Published by Oxford University Press on behalf of the Infectious Diseases Society of America. All rights reserved. For permissions, please e-mail: journals.permissions@oup.com

<https://doi.org/10.1093/cid/ciac299>

24 hours before NF. The large size of this database allowed us to identify 179 matched pairs with versus without NF on the same day after transplantation.

Although the observational design of this study can only be used to define correlations and not causal relationships, we selected a short interval between the sample and subsequent NF to minimize relevant interim events, thus maximizing the chances that any observed associations reflect a directional causal link. First, we found a significant decrease in *Blautia* before NF, without major changes in network characteristics of the gut microbiota. To identify putative mechanisms for this single-taxon effect, we used an independent biorepository of stool and serum samples in patients with acute myeloid leukemia (AML) receiving induction chemotherapy. We found 139 paired samples with serum collected within 24 hours after a stool sample from the same patient. The analysis of stool microbiome and serum metabolome in these sample pairs showed that higher *Blautia* abundances in the gut were associated with higher serum levels of citrulline, a biomarker of intestinal epithelial mass [12, 13], in the next 24 hours.

METHODS

The overall study schema is shown in Figure 1A.

Memorial Sloan Kettering Cancer Center Database

The Memorial Sloan Kettering Cancer Center (MSKCC) HCT data set [14] includes 12 546 longitudinal stool samples collected from 1278 allogeneic HCT recipients. Samples were prospectively collected, most commonly weekly, during the hospitalization for HCT, although some samples were collected before admission and sometimes samples were collected more frequently. In addition to 16S ribosomal RNA amplicon sequences, we used the published “hospitalome” associated with this data set [14] to find patient pairs with NF (cases) and without NF (controls) on the same day relative to transplantation and with a stool sample on the previous day. With the day of transplantation defined as day 0, we explored the database between days −7 and 28 for the occurrence of a first NF episode for each patient. The choice of day −7 was made to capture the beginning of hospitalization in all patients. NF was defined as an oral temperature $\geq 100.4^{\circ}\text{F}$ combined with an absolute neutrophil count $\leq 0.5 \times 10^9/\text{L}$ [15]. We excluded samples with a depth of <5000 reads. This threshold was chosen according to the lower 10% quantile among all samples (4927 reads).

We chose the underlying disease and graft source as criteria for 1:1 matching of cases and controls because they are associated with other important factors that influence the microbiota and NF. As an example, patients with acute leukemia usually receive higher-intensity chemotherapy than patients with lymphoma or myeloma, spend more time in the hospital, and experience a higher antibiotic exposure. Similarly, the

immunosuppressive medications used early after HCT are frequently determined based on the graft source. Therefore, this “bundling” feature is expected to yield a reasonable match for several additional clinical variables. We did not consider antibiotic exposures because although they are a powerful cause of microbiota disruptions, our question concerned the connection between microbiota changes (regardless of their cause) and subsequent NF. After finding 179 pairs of case and control samples matched 1:1 for the underlying disease and graft source in their corresponding patients, we compared cases and controls in aggregate for age, sex, conditioning intensity, and HCT-specific comorbidity index [16]. [Supplementary Data 1](#) provides sample identifiers (IDs) for cases and controls, which may be used to retrieve additional linked data from a prior publication [14].

University of Minnesota Database

The University of Minnesota (UMN) has a large longitudinal gut microbiota data set in patients with AML receiving inpatient induction chemotherapy (National Center for Biotechnology Information sequence read archive no. SRP141394) [10]. This database includes 410 stool samples from 52 patients. A total of 260 serum samples from these patients have been analyzed for metabolomics [10]. We and others have shown that the pattern of microbiota injury in these patients is largely similar to that of allogeneic HCT recipients [4, 17]. Thus, we used this unique microbiome/metabolome database to find potential circulating metabolites that may explain the mechanism of the protective effect of *Blautia* against NF in the MSKCC database. Sample collection started with hospital admission and continued twice weekly (Monday/Thursday ± 2 days) until day 28 of chemotherapy or discharge (whichever occurred first). Serum samples were collected twice weekly (Monday/Thursday; preprandial) between 6 and 8 AM in standard red-top tubes, split in 250- μL aliquots, and stored at -80°C within 2 hours of collection. Stool samples were collected in 95% ethanol-filled sterile tubes and stored at -80°C . We identified 139 same-patient pairs of serum and stool samples, with the stool sample collected within 24 hours before the serum sample. This interval mirrored that used in the analysis of MSKCC data. The short interval also minimized potential intervening confounders and accounted for the short half-life of many circulating metabolites.

Gut Microbiome and Serum Metabolome Profiling

The experimental procedures to generate 16S ribosomal RNA gene amplicon sequences in MSKCC and UMN databases have been described in detail elsewhere [14, 18]. The UMN serum metabolome was profiled using untargeted, ultra-high-performance liquid chromatography tandem mass spectroscopy ([Supplementary Methods](#), [Supplementary Tables 1–4](#), and [Supplementary Figure 1](#)).

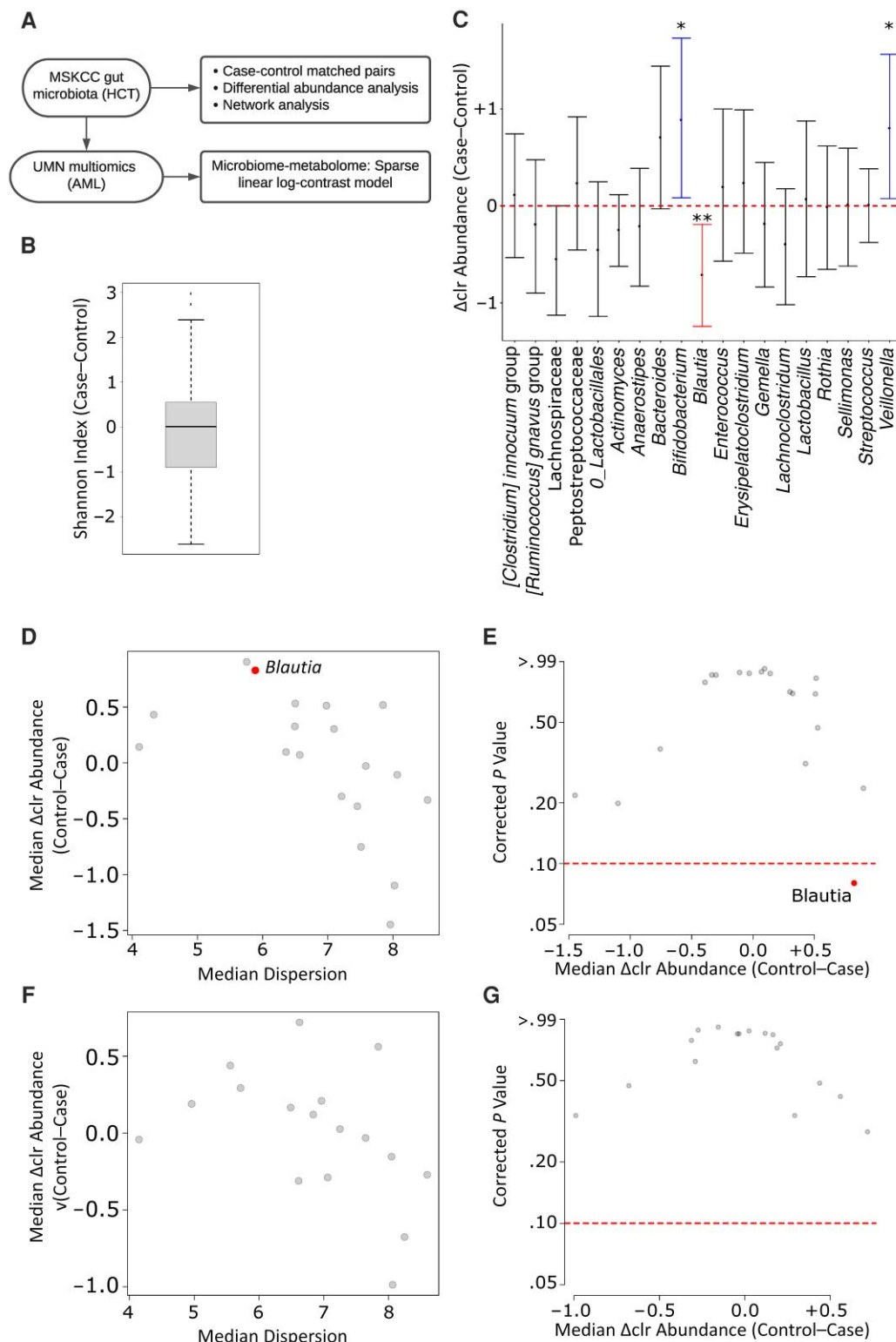


Figure 1. Gut microbiota in cases and controls. *A*, Overall study schema. Abbreviations: AML, acute myeloid leukemia; HCT, hematopoietic cell transplantation; MSKCC, Memorial Sloan Kettering Cancer Center; UMN, University of Minnesota. *B*, Alpha diversity, as measured by the Shannon index. The difference between each case and their matched control was used to generate the plot. The horizontal line in the middle represents the median and the box the interquartile range; the whiskers extend to the last data value that is still within 1.5 times the interquartile range from the lower or upper quartile. *C*, Genus-level difference in centered log-ratio (Δclr) abundance between cases and their matched controls. The y-axis shows median values (with interquartile ranges). Red and blue represent genera that were significantly more abundant in controls or cases, respectively. Significance is shown by one ($P < .05$) or two ($P < .01$) asterisks. No formal differential abundance analysis was performed in this plot. *D–G*, ALDEx2 results with a 1-day (*D*, *E*) versus 2-day (*F*, *G*) window interval between the stool sample and subsequent neutropenic fever (cases) or no neutropenic fever (controls). Differentially abundant genera ($q < 0.10$ in *aldex.test*) are shown in red.

Statistical Analysis

Methodological details are provided in [Supplementary Methods](#). A paired Wilcoxon test was used to compare MSKCC cases and controls for alpha diversity. Differential abundance analysis using ALDEx2 [19] was performed to identify overrepresented and underrepresented taxa in each group. The MSKCC gut microbiota network was compared between the groups using the 100 most abundant genera (NetCoMi package [20]). To evaluate the potential mechanisms by which differentially abundant taxa may protect against or promote NF, a multiomics analysis of the UMN gut microbiome and serum metabolome database was performed, using sparse linear log-contrast modeling [21].

RESULTS

A total of 179 samples from 179 unique case and their matched paired 179 samples from 117 unique controls in the MSKCC database were analyzed ([Table 1](#)). Case samples were individually matched to control samples for the underlying disease and graft source. The underlying disease distribution included leukemia in 98 pairs (55%), myeloproliferative neoplasm or myelodysplastic syndrome in 41 (23%), lymphoma in 24 (13%), and multiple myeloma in 16 (9%). Graft source distribution included bone marrow in 23 pairs (13%), peripheral blood in 47 (26%), cord blood in 18 (10%), and T-deplete grafts in 91 (51%). When the included patients were considered for other important clinical variables in aggregate, there were no significant differences between the groups in sex, conditioning intensity, and HCT-specific comorbidity index. Controls were significantly older than cases (mean age, 56 vs 53 years, respectively; $P=.04$), but this difference did not seem clinically meaningful. Infection workup at the time of NF in cases revealed *Escherichia coli* in 8 patients, vancomycin-resistant enterococci in 2, and *Enterobacter* species in 2. The median (range) number of reads per sample was 28 428 (range, 5021–703 665). Alpha diversity was similar in both groups (median, 2.60 in cases vs 2.54 in controls, respectively; median difference, 0.009; paired Wilcoxon $P=.35$; [Figure 1B](#)).

When microbiota composition in the 2 groups was compared using a simple analysis of pairwise differences in genus-level abundances, *Blautia* was significantly more abundant in controls (paired Wilcoxon $P=.007$), while *Bifidobacterium* and *Veillonella* (both $P=.03$) were more abundant in cases ([Figure 1C](#)). Next, we proceeded to formal differential abundance analysis using the 19 postfiltering genera ([Supplementary Data 2](#)). *Blautia* was the only differentially abundant genus, significantly more abundant in controls (corrected paired Wilcoxon $P=.08$; [Figure 1D](#) and [1E](#)). In exploratory analysis with a 2-day interval allowed between the stool sample and fever and 294 eligible pairs of matched case and control samples, no differentially abundant genera were

found ([Figure 1F](#) and [1G](#)), indicating that the apparent protective effect of *Blautia* occurs on a fast time scale.

Next, we asked whether the relative depletion of *Blautia* before NF represents a single-taxon effect or an indicator of a broader, community-level change in the gut microbiota. In comparative network analysis of the microbiota in cases and controls, we found no significant differences between the groups in any of the studied measures ([Figure 2](#) and [Supplementary Data 3](#)). The *Blautia* cluster was highly similar in the 2 groups and included obligate anaerobic commensals in the Lachnospiraceae (*Lachnoclostridium* [*Ruminococcus*] *gnavus* group, [*Ruminococcus*] *torques* group, *Sellimonas*, *Anaerostipes*) and Oscillospiraceae (*UBA1819*, *Ruminiclostridium* 5) families ([Figure 2](#)). The strongest association of *Blautia* in both groups was with *Lachnoclostridium* ([Figure 2](#)). These findings suggested that the negative association between *Blautia* and NF was a single-taxon effect rather than a reflection of a broader community-level change in the gut microbiota.

We then evaluated antibiotic exposures as a potential cause for *Blautia* loss. We defined each sample as *Blautia* low or high based on its relative abundance of *Blautia* compared with the median *Blautia* abundance in all samples (12.6%). We considered all antibacterial antibiotic exposures before NF in cases and before the same day relative to HCT in their matched controls. We found antibiotic exposures among *Blautia*-low samples to be consistently heavier than in *Blautia*-high samples ([Figure 3](#)). This suggests that antibiotic exposure was a major cause of *Blautia* loss.

To explore potential mechanisms by which *Blautia* may protect against NF, we investigated the relationship between the gut microbiota and serum metabolome in an independent multiomics database from the UMN. Although the patients in this database were enrolled at the time of induction chemotherapy, their patterns of gut microbiota changes largely resemble those in allogeneic HCT recipients [4]. The 2 populations have many microbiome-relevant similarities, including mucositis due to cytotoxic chemotherapy, nutritional compromise, prolonged hospital stay, and extensive healthcare and antibiotic exposure. However, there are also some important differences between the 2 patient populations. Most HCT recipients have previously experienced microbiome damage due to their prior treatments, whereas many patients with AML have new diagnoses, without a significant prior history. The presence of an allogeneic immune system and the use of total body irradiation are other unique features of HCT recipients. We included data from patients with AML with the goal of finding novel insight into a plausible mechanism for the observed association in HCT recipients, while understanding that confident extrapolation may not be possible given the differences between the 2 patient populations.

The gut microbiota is known to regulate and significantly contribute to the circulating metabolome [22–24]. Thus, we

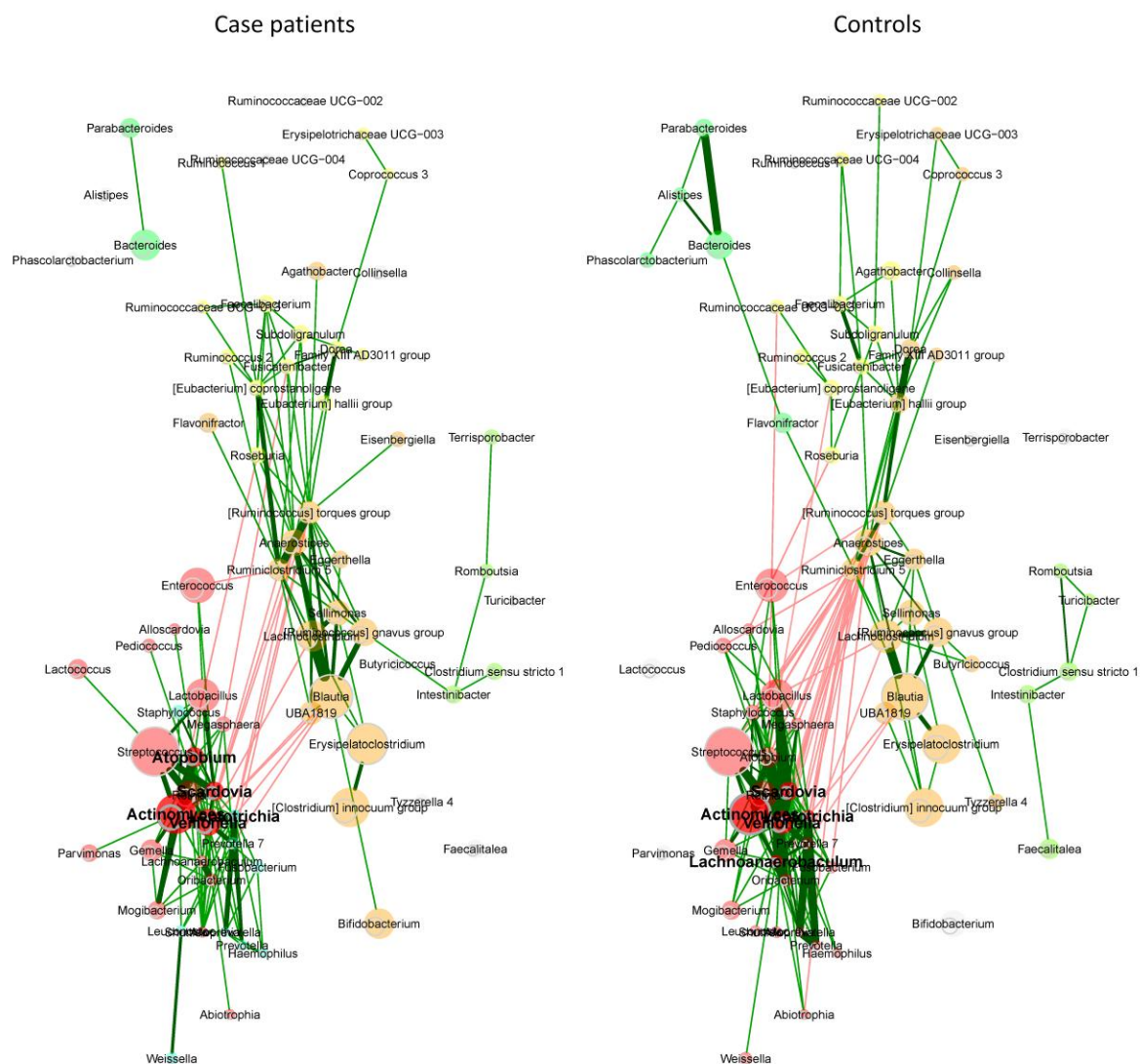


Figure 2. Network analysis of the gut microbiota. Two clusters have the same color in both groups if they have ≥ 2 nodes in common. Node sizes are scaled to the column sums of centered log-ratio-transformed data. Edge weights are (nonnegative) similarities, with red and green representing negative and positive associations, respectively.

used this unique multiomics database to address the question of interest in the present study. In all, 139 same-patient stool-serum sample pairs from 51 patients were identified and analyzed by means of sparse linear log-contrast modeling. Each stool sample was collected within 24 hours before its corresponding serum sample. Table 2 summarizes patient characteristics. We detected 945 circulating metabolites, 872 of which were detectable in at least half of the samples (Supplementary Data 4). The 10 postfiltering genera were included in the model as potential predictors of the 872 metabolites.

Although each genus was associated with several metabolites (Figure 4 and Supplementary Data 5), *Blautia* was the only genus that showed exclusively positive associations (Supplementary Data 6), suggesting its unique

metabolic relevance. Citrulline was one of 21 associated metabolites associated with *Blautia*. To exclude the possibility of this relationship being driven primarily by patient-level factors owing to the longitudinal nature of the database, we built a mixed-effects model (lme4 package) in the form of $Blautia \sim \text{patient ID} + \text{citrulline}$, where patient ID was considered a random effect. After controlling for patient ID, *Blautia* remained a significant predictor of citrulline (regression coefficient, 6.01 [95% confidence interval, .16–10.90]; $P = .04$). A conventional univariate correlation analysis showed a significant ($P = .04$) but weak (Pearson $R = 0.17$) relationship between *Blautia* and citrulline, indicating the presence of additional regulators of enterocyte mass.

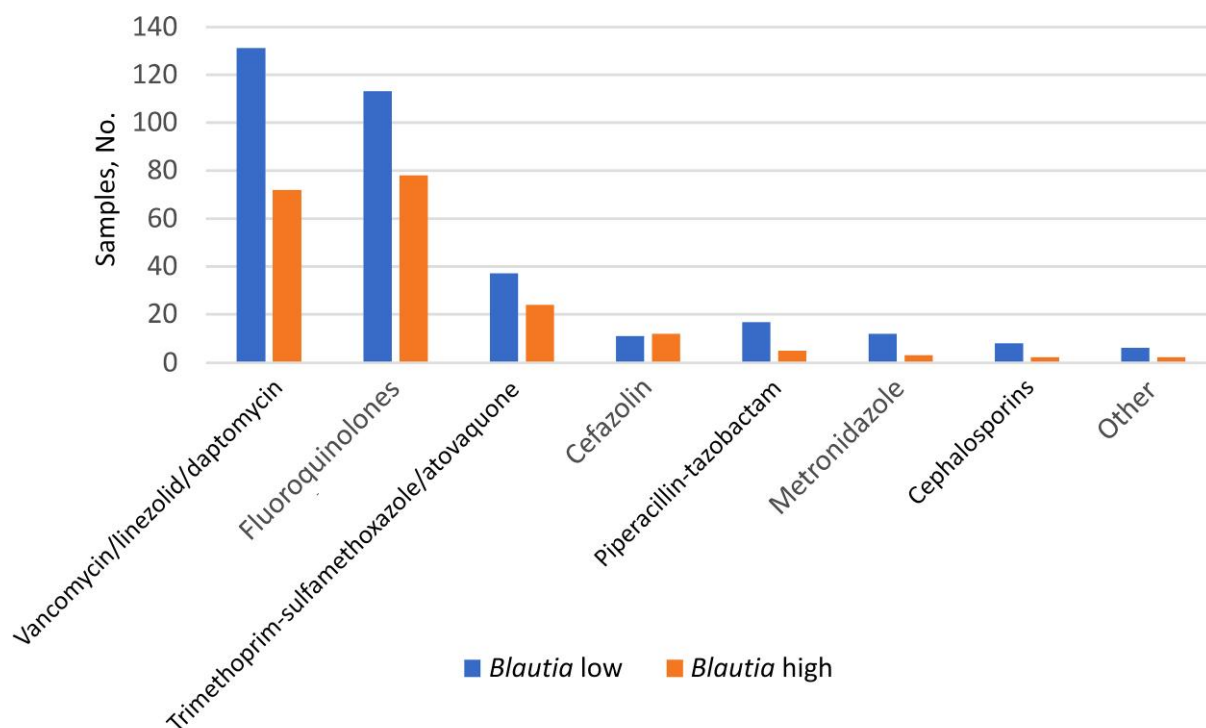


Figure 3. Antibacterial antibiotic exposures before samples with low versus high *Blautia* abundance. Low and high *Blautia* abundance were defined based on whether the relative abundance of *Blautia* in the sample was lower or higher than the median among all samples. Exposure was defined binarily. Cephalosporins were third generation or higher.

DISCUSSION

Using a large HCT microbiota database with granular hospital-ome and an independent, unique multiomics database in patients with AML, we found evidence for a single-taxon association between *Blautia* and NF, possibly through strengthening of the intestinal epithelium. Because HCT recipients are

severely immunocompromised and at high risk for bleeding and infection early after HCT, mechanistic studies are usually infeasible, so we considered the approach taken here to be the next best alternative.

In one of the earliest microbiota studies in HCT recipients, intestinal *Blautia* was associated with a reduced mortality rate from acute graft-vs-host disease and improved overall survival [25]. This is consistent with the known anti-inflammatory effects of *Blautia* [26, 27]. Other metabolic activities of *Blautia*

Table 1. Patient Characteristics in the Memorial Sloan Kettering Cancer Center Cohort

Characteristic	Patients, No. (%) ^a		P-value
	Cases (n = 179)	Controls (n = 117)	
Age, mean (SD), y	53 (13)	56 (12)	.04
Sex			
Male	120 (67)	70 (60)	.25
Female	59 (33)	47 (40)	
HCT-CI			
0–1	54 (31)	36 (32)	>.99
≥2	119 (69)	78 (68)	
Conditioning intensity			
Myeloablative	124 (69)	69 (59)	.11
Reduced intensity	11 (6)	6 (5)	
Nonmyeloablative	44 (25)	42 (36)	

Abbreviation: HCT-CI, HCT-specific comorbidity index; SD, standard deviation.

^aData represent no. (%) of patients unless otherwise specified.

Table 2. Patient Characteristics in the University of Minnesota Cohort

Characteristic	Patients, No. (%) ^a (n = 51)
Age, mean (SD), y	57 (15)
Sex	
Male	30 (59)
Female	21 (41)
Disease phase	
Newly diagnosed	44 (86)
Relapsed	7 (14)
Chemotherapy regimen	
7 + 3 (Conventional or liposomal)	35 (69)
Clofarabine + cytarabine	9 (18)
Venetoclax + hypomethylator	3 (6)
Other	4 (8)

^aData represent no. (%) of patients unless otherwise specified.

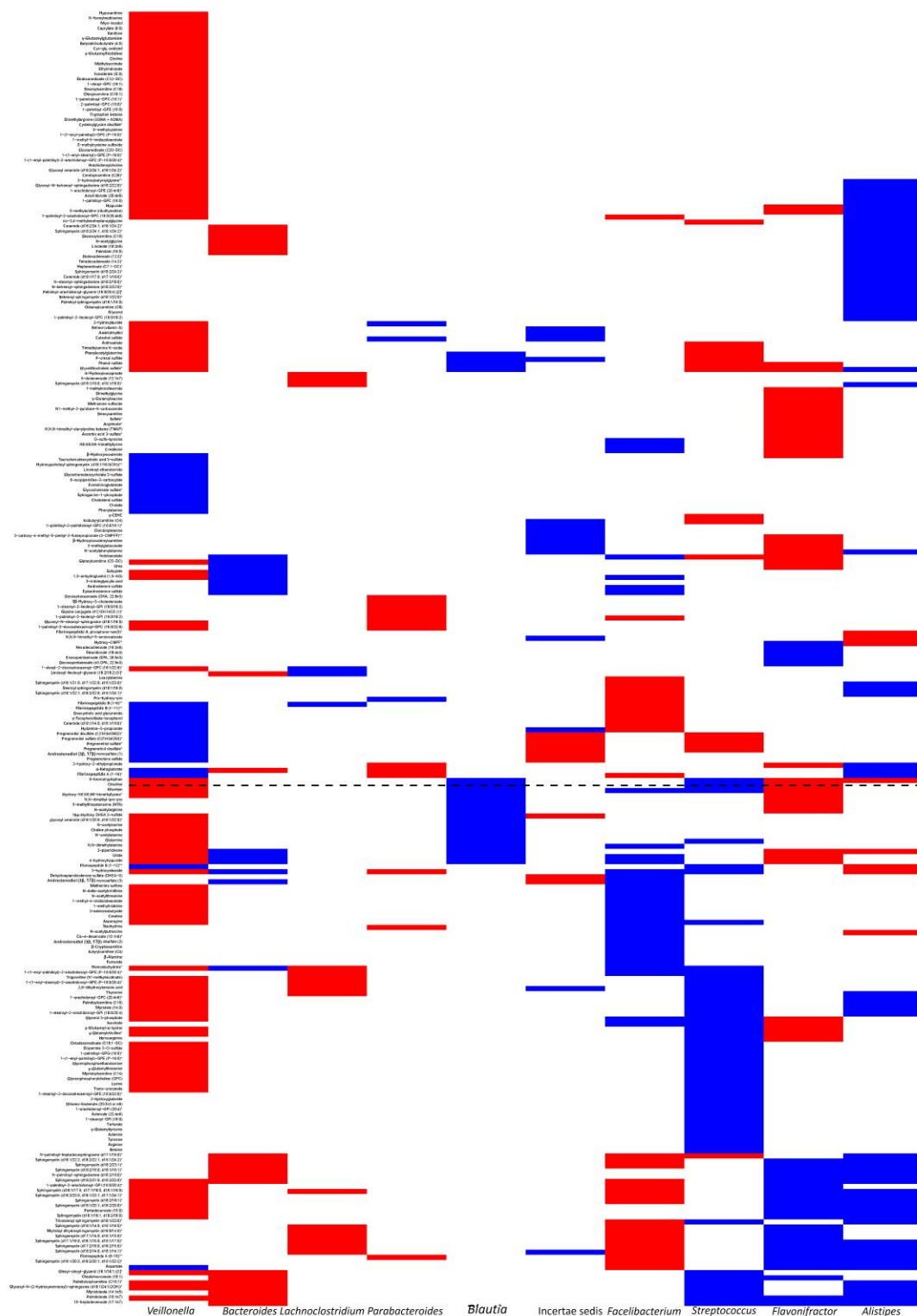


Figure 4. Gut microbiome predictors of next-day metabolome. Results of sparse linear log-contrast modeling using the gut microbiota as predictors of next-day serum metabolome in patients with acute myeloid leukemia. The genera along the x-axis are those that remained in ≥ 90 of the 100 bootstraps and showed a uniformly positive or negative association with ≥ 1 metabolite. Red and blue represent negative and positive associations, respectively. Associations with $< 90\%$ consistency across the bootstraps are not colored. *Blautia* is shown in a larger font, and the dashed horizontal line indicates citrulline. This plot is provided in tabular format in [Supplementary Data 4](#) for better visibility of metabolite names.

include bacteriocin production, which can modulate the gut microbiota and prevent pathogen colonization [28, 29]. Our network analysis did not find any community-level changes in the gut microbiota before NF, suggesting a single-taxon phenomenon. Our metabolomic analysis found a strong association between *Blautia* and circulating citrulline. Citrulline is an amino acid produced exclusively by intestinal epithelial cells [30], with circulating levels indicating total functioning enterocyte mass. Citrulline has been established as a biomarker for gut barrier integrity and function [30, 31], and its low levels in the blood are associated with bacteremia and inflammation [32].

Induction chemotherapy and conditioning regimens used for HCT cause cytotoxic damage to the intestinal epithelium, which may be partially mitigated by the protective microbiota, such as *Blautia*. Considering the primacy of gut barrier damage in the pathogenesis of acute intestinal graft-vs-host disease [33], this model provides an additional mechanism by which *Blautia* may protect against fatal intestinal graft-vs-host disease. Our group's previous work suggested that the expansion of the mucolytic genus *Akkermansia* may increase the risk of NF through gut barrier modulation [10, 11]. However, while the association between *Blautia* abundance and freedom from NF was observable in immediately adjacent time points (within 24 hours), *Akkermansia* associations have been observed over a time span of 5–7 days, suggesting different modes of action on the epithelium. The lack of an association between *Blautia* and *Akkermansia* abundances in our network analysis does not support a functional relationship between the 2 genera, though this possibility cannot be excluded.

The current study has some limitations. Because amplicon data do not have reliable species-level resolution [34], we do not know which species of *Blautia* are involved. In addition, because network analysis was based on the most abundant taxa, subtle community structure differences involving less abundant taxa may have been present between the groups that could not be detected. Citrulline was only one of the metabolites associated with *Blautia*. We found several other metabolites that could protect against NF, including those involved in amino acid, secondary bile acid, and lipid metabolism. Short-chain fatty acids such as butyrate, produced from dietary fiber by several predominantly anaerobic members of the gut microbiome, have been associated with anti-inflammatory effects through an inhibitory effect on histone deacetylase-mediated pathways and, through metabolism to acetyl coenzyme A, as an enterocyte tricarboxylic acid cycle-mediated energy source, thereby contributing to enterocyte health and mucosal integrity [35]. Although we found no significant association between *Blautia* and short-chain fatty acids, this possibility is worth further exploration. Finally, the extent to which data from the AML cohort might help explain the mechanisms implicated in the HCT cohort are unclear at this time, and any extrapolation should be treated with caution.

In conclusion, we propose that NF-relevant effects of the gut microbiota on the intestinal epithelium occur on 2 time scales. A protective effect, perhaps mediated by *Blautia*, might occur on the faster time scale, where *Blautia* or its products modulate the health of the epithelium. The absence of an association between *Blautia* and NF when the interval was increased from 1 to 2 days could indicate a short-lasting effect by a *Blautia* metabolite with a short half-life. In contrast, the detrimental effect mediated by *Akkermansia* occurs on the slower time scale where *Akkermansia* degrades the mucus and thins the intestinal barrier. As an obligate anaerobic commensal of the gut, *Blautia* depletion has been associated with the use of anti-anaerobic antibiotics and parenteral nutrition [25]. Similarly, dietary fiber deprivation in mice caused expansion of *Akkermansia* [36]. Therefore, restricted use of anti-anaerobic antibiotics and resumption of fiber-rich oral intake as early as possible could be effective strategies to reduce NF. Selective restoration of *Blautia* to sustain high levels during the periods of neutropenia in patients at high risk for depletion is another potential strategy for future trials. The precise mechanism by which *Blautia* may augment intestinal health is unknown and in need of further research.

Supplementary Data

Supplementary materials are available at *Clinical Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

Author contributions. A. R., J. U. P., and C. S. conceived the study and performed the analyses. A. R., J. U. P., M. E., T. U. R., H. E., and L. T. M. collected metadata. H. H., T. K., and C. S. extracted and sequenced DNA for the University of Minnesota database. A. R. wrote the manuscript. J. U. P., M. E., S. G. H., A. K., D. J. W., and C. S. provided critical feedback on the manuscript.

Acknowledgments. Sequence data from University of Minnesota samples were analyzed using the resources of the Minnesota Supercomputing Institute. Serum samples were analyzed for metabolomics by Metabolon.

Disclaimer. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health's National Center for Advancing Translational Sciences.

Financial support. This work was supported by the National Center for Advancing Translational Sciences, National Institutes of Health (NIH) (grants KL2TR002492 and UL1TR002494 to the University of Minnesota), NHLBI, NIH (grant K08HL143189 to J. U. P.), the National Cancer Institute, NIH (core grant NCI P30 CA008748 to Memorial Sloan Kettering Cancer Center [MSKCC]), and the Parker Institute for Cancer Immunotherapy at MSKCC (of which J. U. P. is a member).

Potential conflicts of interest. J. U. P. reports research funding, intellectual property fees, and travel reimbursement from Seres Therapeutics and consulting fees from DaVolterra, CSL Behring, MaaT Pharma, Arjun Pai, and the French National Cancer Institute. He serves on an advisory board for and holds equity in Postbiotics Plus Research. He has filed intellectual property applications related to the microbiome (reference nos. 62/843,849, 62/977,908, and 15/756,845). MSKCC has financial interests relative to Seres Therapeutics. J. U. P. reports the grants or contracts from the following: Society for MSKCC (research grant), Adult BMT Service Translational Research Award (Pilot Grant), Seres Therapeutics (sponsored research),

Starr Cancer Consortium, Cycle for Survival Foundation, the National Cancer Institute (2 P01 CA023766 39A1), and the Memorial Sloan Kettering/Hackensack Meridian Health Partnership Immunology Research Immunology Research Collaboration. J. U. P. also reports licensing fees from Seres Therapeutics; payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing, or educational events from the University of Nebraska Medical Center, HayMatic Meetings & Events, the Japanese Society of Hematology, William Blair & Co, KongresKompagniet, the Clinical Immunology Society, the University of Pennsylvania, University Hospital Regensburg, and Hanson Wade; and support for attending meetings and/or travel from the Clinical Immunology Society, the University of Nebraska Medical Center, the University of Pennsylvania, and Parker Institute for Cancer Immunotherapy. J. U. P. asserts 2 patents planned, issued, or pending for β -lactamase compositions for treatment of graft-vs-host disease, and bacterial compositions and methods for cancer survival. Finally, J. U. P. holds stock or stock options with Postbiotics Plus Research. All other authors report no potential conflicts. All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

References

- Rearigh L, Stohs E, Freifeld A, Zimmer A. De-escalation of empiric broad spectrum antibiotics in hematopoietic stem cell transplant recipients with febrile neutropenia. *Ann Hematol* **2020**; 99:1917–24.
- Celebi H, Akan H, Akçağlayan E, Ustün C, Arat M. Febrile neutropenia in allogeneic and autologous peripheral blood stem cell transplantation and conventional chemotherapy for malignancies. *Bone Marrow Transplant* **2000**; 26:211–4.
- Holler E, Butzhammer P, Schmid K. Metagenomic analysis of the stool microbiome in patients receiving allogeneic stem cell transplantation: loss of diversity is associated with use of systemic antibiotics and more pronounced in gastrointestinal graft-versus-host disease. *Biol Blood Marrow Transplant* **2014**; 20:640–5.
- Rashidi A, Kaiser T, Graiziger C, et al. Gut dysbiosis during antileukemia chemotherapy versus allogeneic hematopoietic cell transplantation. *Cancer* **2020**; 126: 1434–47.
- Peled JU, Gomes ALC, Devlin SM, et al. Microbiota as predictor of mortality in allogeneic hematopoietic-cell transplantation. *N Engl J Med* **2020**; 382:822–34.
- Berg RD. Bacterial translocation from the gastrointestinal tracts of mice receiving immunosuppressive chemotherapeutic agents. *Curr Microbiol* **1983**; 8:285–92.
- Tamburini FB, Andermann TM, Tkachenko E, Senchyna F, Banaei N, Bhatt AS. Precision identification of diverse bloodstream pathogens in the gut microbiome. *Nat Med* **2018**; 24:1809–14.
- Taur Y, Xavier JB, Lipuma L, et al. Intestinal domination and the risk of bacteremia in patients undergoing allogeneic hematopoietic stem cell transplantation. *Clin Infect Dis* **2012**; 55:905–14.
- Stoma I, Littmann ER, Peled JU, et al. Compositional flux within the intestinal microbiota and risk for bloodstream infection with Gram-negative bacteria. *Clin Infect Dis* **2021**; 73:e4627–35.
- Rashidi A, Ebadi M, Rehman TU, et al. Altered microbiota-host metabolic cross talk preceding neutropenic fever in patients with acute leukemia. *Blood Adv* **2021**; 5:3937–50.
- Rashidi A, Kaiser T, Graiziger C, et al. Specific gut microbiota changes heralding bloodstream infection and neutropenic fever during intensive chemotherapy. *Leukemia* **2020**; 34:312–6.
- Derikx JPM, Blijlevens NMA, Donnelly JP, et al. Loss of enterocyte mass is accompanied by diminished turnover of enterocytes after myeloablative therapy in hematopoietic stem-cell transplant recipients. *Ann Oncol* **2009**; 20:337–42.
- Herbers AHE, Blijlevens NMA, Donnelly JP, de Witte TJM. Bacteraemia coincides with low citrulline concentrations after high-dose melphalan in autologous HSCT recipients. *Bone Marrow Transplant* **2008**; 42:345–9.
- Liao C, Taylor BP, Ceccarani C, et al. Compilation of longitudinal microbiota data and hospitalome from hematopoietic cell transplantation patients. *Sci Data* **2021**; 8:71.
- Freifeld AG, Bow EJ, Sepkowitz KA, et al. Clinical practice guideline for the use of antimicrobial agents in neutropenic patients with cancer: 2010 update by the Infectious Diseases Society of America. *Clin Infect Dis* **2011**; 52:e56–93.
- Sorror ML, Maris MB, Storb R, et al. Hematopoietic cell transplantation (HCT)-specific comorbidity index: a new tool for risk assessment before allogeneic HCT. *Blood* **2005**; 106:2912–9.
- Galloway-Peña J, Brumlow C, Shelburne S. Impact of the microbiota on bacterial infections during cancer treatment. *Trends Microbiol* **2017**; 25:992–1004.
- Rashidi A, Ebadi M, Rehman TU, et al. Gut microbiota response to antibiotics is personalized and depends on baseline microbiota. *Microbiome* **2021**; 9:211.
- Fernandes AD, Reid JNS, Macklaim JM, McMurrough TA, Edgell DR, Gloor GB. Unifying the analysis of high-throughput sequencing datasets: characterizing RNA-seq, 16S rRNA gene sequencing and selective growth experiments by compositional data analysis. *Microbiome* **2014**; 2:15.
- Peschel S, Müller CL, von Mutius E, Boulesteix AL, Depner M. NetCoMi: network construction and comparison for microbiome data in R. *Brief Bioinform* **2021**; 22: bbaa290.
- Lin W, Shi P, Feng R, Li H. Variable selection in regression with compositional covariates. *Biometrika* **2014**; 101:785–97.
- Wikoff WR, Anfora AT, Liu J, et al. Metabolomics analysis reveals large effects of gut microflora on mammalian blood metabolites. *Proc Natl Acad Sci USA* **2009**; 106:3698–703.
- Visconti A, Le Roy CI, Rosa F, et al. Interplay between the human gut microbiome and host metabolism. *Nat Commun* **2019**; 10:4505.
- Bar N, Korem T, Weissbrod O, et al. A reference map of potential determinants for the human serum metabolome. *Nature* **2020**; 588:135–40.
- Jenq RR, Taur Y, Devlin SM, et al. Intestinal blautia is associated with reduced death from graft-versus-host disease. *Biol Blood Marrow Transplant* **2015**; 21: 1373–83.
- Chen W, Liu F, Ling Z, Tong X, Xiang C. Human intestinal lumen and mucosa-associated microbiota in patients with colorectal cancer. *PLoS One* **2012**; 7:e39743.
- Bajaj JS, Hylemon PB, Ridlon JM, et al. Colonic mucosal microbiome differs from stool microbiome in cirrhosis and hepatic encephalopathy and is linked to cognition and inflammation. *Am J Physiol Gastrointest Liver Physiol* **2012**; 303: G675–85.
- Kim SG, Becattini S, Moody TU, et al. Microbiota-derived lantibiotic restores resistance against vancomycin-resistant *Enterococcus*. *Nature* **2019**; 572:665–9.
- Caballero S, Kim S, Carter RA, et al. Cooperating commensals restore colonization resistance to vancomycin-resistant *Enterococcus faecium*. *Cell Host Microbe* **2017**; 21:592–602.e4.
- Demacker PNM, Beijers AM, van Daal H, Peter Donnelly J, Blijlevens NMA, van den Ouweland JMW. Plasma citrulline measurement using UPLC tandem mass-spectrometry to determine small intestinal enterocyte pathology. *J Chromatogr B* **2009**; 877:387–92.
- Hueso T, Ekpe K, Mayeur C, et al. Impact and consequences of intensive chemotherapy on intestinal barrier and microbiota in acute myeloid leukemia: the role of mucosal strengthening. *Gut Microbes* **2020**; 12:1800897.
- van der Velden WJFM, Herbers AHE, Netea MG, Blijlevens NMA. Mucosal barrier injury, fever and infection in neutropenic patients with cancer: introducing the paradigm febrile mucositis. *Br J Haematol* **2014**; 167:441–52.
- Hill GR, Ferrara JLM. The primacy of the gastrointestinal tract as a target organ of acute graft-versus-host disease: rationale for the use of cytokine shields in allogeneic bone marrow transplantation. *Blood*, *J Am Soc Hematol* **2000**; 95:2754–9.
- Johnson JS, Spakowicz DJ, Hong B-Y, et al. Evaluation of 16S rRNA gene sequencing for species and strain-level microbiome analysis. *Nat Commun* **2019**; 10:5029.
- Donohoe DR, Garge N, Zhang X, et al. The microbiome and butyrate regulate energy metabolism and autophagy in the mammalian colon. *Cell Metab* **2011**; 13: 517–26.
- Desai MS, Seekatz AM, Koropatkin NM, et al. A dietary fiber-deprived gut microbiota degrades the colonic mucus barrier and enhances pathogen susceptibility. *Cell* **2016**; 167:1339–53.e21.